Marco Baj

Via Cavour, 1, 21041, Albizzate (VA), Italy
marcobaj24@gmail.com

PERSONAL PROFILE

Recent Master's graduate in Space Engineering with a strong foundation in dynamical systems and computational modeling. Driven by deep research interests in astrodynamics, celestial mechanics, and planetary science, I am seeking a PhD position in a stimulating research environment where I can apply my skills to Solar System dynamics and spaceflight applications.

EDUCATION

Master of Science, Space Engineering Sep 2023 – Dec 2025
Politecnico di Milano
Final Grade: 110/110 *cum laude*

- **Relevant Courses:** Space Physics, Orbital Mechanics, Modeling and Simulation of Aerospace Systems, Spacecraft Attitude Dynamics, Spacecraft Guidance and Navigation.
- **Thesis:** *Low-energy dynamics around Pan and ring particle accretion as the origin of its equatorial ridge*
  Supervisors: Prof. Iosto Fodde, Prof. Fabio Ferrari, Lucia Francesca Civati
  - Executed high-fidelity numerical simulations of particle accretion using perturbed gravitational dynamics and large-scale grid-search trajectory propagation.
  - Conducted statistical analysis of impact distributions and characteristics.
  - Showed that low-energy ring particle accretion under tidally locked conditions can reproduce key features of Pan's equatorial ridge morphology.

Bachelor of Science, Aerospace Engineering Sep 2020 – Sep 2023
Politecnico di Milano
Final Grade: 103/110

- **Relevant Courses:** Applied Numerical Analysis, Aerospace Mechanics, Physics of Waves, Dynamics of Aerospace Systems, Fluid Dynamics.

RESEARCH

Submitted Manuscripts

- Baj, M., Fodde, I., Civati, L. F., and Ferrari, F. *Low-energy ring particle accretion as the origin of Pan's equatorial ridge*, submitted to Astronomy & Astrophysics, preprint available at arxiv.org/abs/2609.03060.

Conference Contributions

- Baj, M., Fodde, I., Civati, L. F., and Ferrari, F. *The Shape of Small Embedded Ring Moonlets: Low-Energy Dynamics and the Origin of Pan's Polygonal Ridge*, Europlanet Science Congress 2026, The Hague, The Netherlands, 7–11 Sep 2026, EPSC2026-778, doi.org/10.5194/epsc2026-778.

TECHNICAL SKILLS

- **Programming & Scientific Computing:** MATLAB, C/C++, CUDA C++ (GPU Programming).

- **Astrodynamics Tools:** SPICE Toolkits, NASA JPL Horizons System, Two-Line Element (TLE) processing and propagation.
- **Simulation Software:** Simulink, Simscape, Dymola.
- **CAD & Prototyping:** SolidWorks, Autodesk Inventor, Solid Edge, Fusion 360, Arduino, Raspberry Pi, 3D Printing.

ADDITIONAL TRAINING & WORKSHOPS

- **Workshop – Human Factors and Teamwork in Spaceflight Operations** Mar 2025
  European Space Agency (ESA)
- **Practical Course of Scientific Communication** Dec 2023
  Politecnico di Milano

OUTREACH ACTIVITIES

- **International Astronautical Congress (IAC)** Oct 2024
  Assisted the COMPASS Research Group during public engagement operations.
- **Festival dell'Ingegneria** Sep 2024
  Supported the COMPASS Research Group's exhibition activities at the 2024 event.
- **Educational Podcast** Nov 2023
  Produced an educational podcast episode focusing on space debris mitigation.
- **Festival dell'Ingegneria** Sep 2023
  Supported the COMPASS Research Group's exhibition activities at the 2023 event.

AREAS OF INTEREST

- Modeling and Numerical Simulation
- Planetary and Small Bodies Science
- Astrodynamics and Celestial Mechanics
- Solar System Dynamics
- Space Mission Design

LANGUAGES

- **Italian:** Native
- **English:** Fluent